

# **GEOGRAPHY**

# **PHYSICAL GEOGRAPHY**



## **GEOMORPHOLOGY**

### **MODULE - 37**

#### **DESERT TOPOGRAPHY-2**

- **EROSIONAL LANDFORMS**
- **DEPOSITIONAL LANDFORMS**



# ARID TOPOGRAPHY

## EROSIONAL LANDFORMS

6

### PEDIMENTS AND PEDIPLAINS

- 1** Landscape evolution in deserts is primarily concerned with the formation and extension of pediments.



- 2** Gently inclined rocky floors close to the mountains at their foot with or without a thin cover of debris, are called pediments.

- 3** Such rocky floors form through the erosion of mountain front through a combination of lateral erosion by streams and sheet flooding.

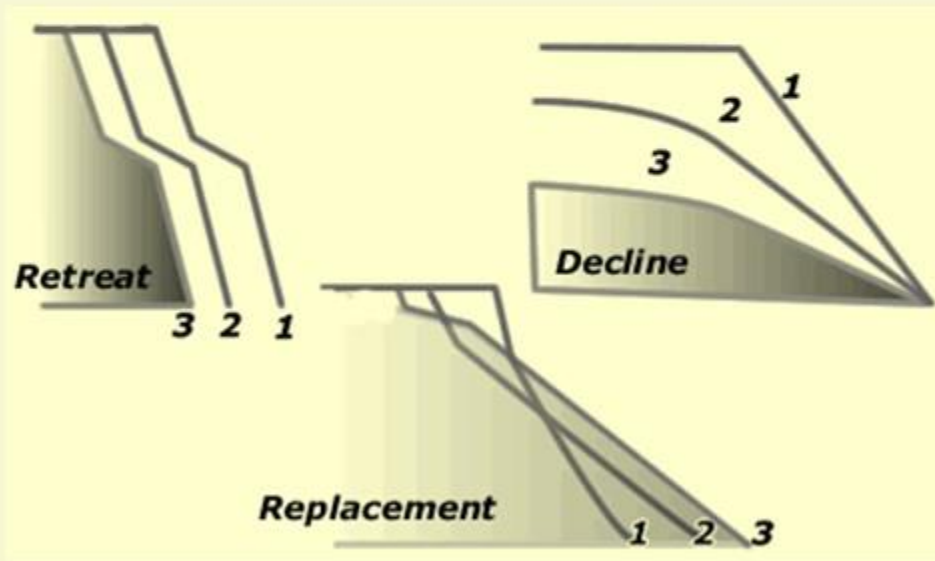
- 4** Erosion starts along the steep margins of the landmass or the steep sides of the tectonically controlled steep incision features over the landmass.

# ARID TOPOGRAPHY

## EROSIONAL LANDFORMS

### 6 PEDIMENTS AND PEDIPLAINS

- 5** Once, pediments are formed with a steep wash slope followed by cliff or free face above it, the steep wash slope and free face retreat backwards.



- 6** This method of erosion is termed as **parallel retreat of slopes through backwasting**.

- 7** So, through parallel retreat of slopes, the pediments extend backwards at the expense of mountain front.

- 8** Gradually, the mountain gets **reduced leaving an inselberg which is a remnant of the mountain**.

- 9** That's how the **high relief in desert areas is reduced to low featureless plains called pediplains**.



# ARID TOPOGRAPHY

## EROSIONAL LANDFORMS

7

### PLAYAS



- 1 In arid areas, the outflowing streams from the upland regions are both short and intermittent.
- 2 They drain into the lower depressions so that drainage is almost entirely internal.
- 3 Sometimes water collected in a depression or a desert basin does not completely disappear by evaporation and a temporary lake is formed.

# ARID TOPOGRAPHY

## EROSIONAL LANDFORMS

7

### PLAYAS



4 Such lakes contain a high percentage of salts, because of comparatively high evaporation and are glistening white when they dry up.

5 The lakes and alluvial plains formed by them are called **playas** or **Salinas**.



# ARID TOPOGRAPHY

## EROSIONAL LANDFORMS

### 8 DEFLATION HOLLOWES & CAVES



- 1 Weathered mantle from over the rocks or bare soil, gets blown out by persistent movement of wind currents in one direction.
- 2 This process may create shallow depressions called deflation hollows.
- 3 Deflation also creates numerous small pits or cavities over rock surfaces.
- 4 The rock faces suffer impact and abrasion of wind-borne sand and first shallow depressions called blow outs are created and some of the blow outs become deeper and wider, fit to be called as caves.

# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

- 1** Wind is a good sorting agent. Depending upon the velocity of wind, **different sizes of grains are moved along the floors by rolling or saltation and carried in suspension** and in this process of transportation itself, the materials get sorted.
- 2** When the wind slows or begins to die down, **depending upon sizes of grains and their critical velocities**, the grains will begin to settle.
- 3** So, in depositional landforms made by wind, good sorting of grains can be found.
- 4** Since wind is there everywhere and wherever there is good source of sand and with constant wind directions, depositional features in arid regions can develop anywhere.



# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

1

### SAND DUNES



1

Dry hot deserts are good places for sand dune formation.

2

Obstacles to initiate dune formation are equally important. There can be a great variety of dune forms



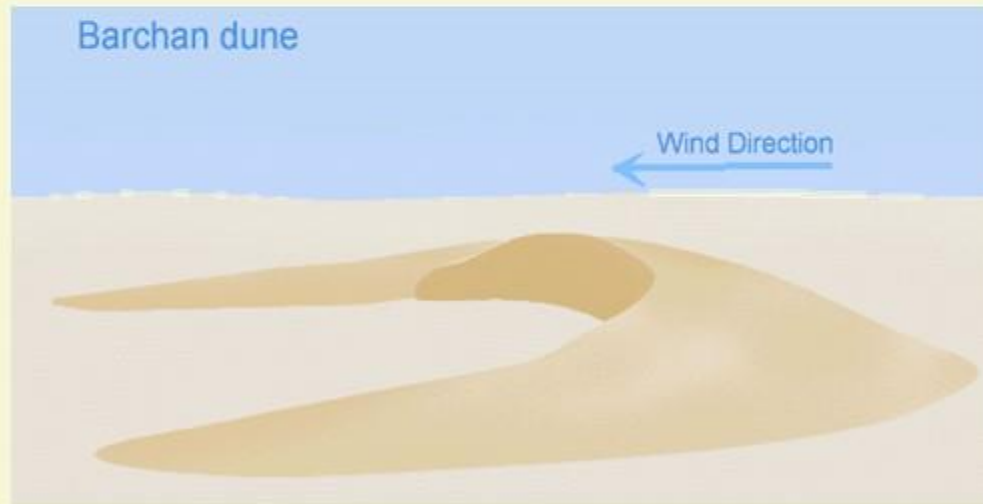
# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

2

### BARCHANS

- 1 Crescent shaped dunes called Barchans with the points or wings directed away from wind direction i.e. downwind form where the wind direction is constant and moderate and where the original surface over which sand is moving is almost uniform.



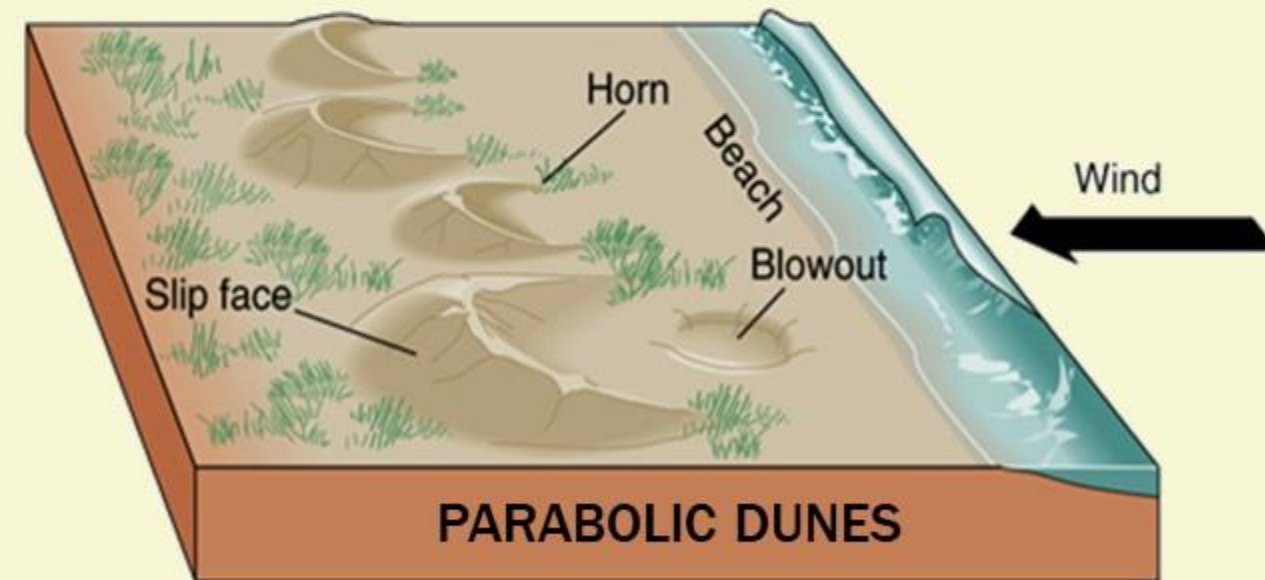
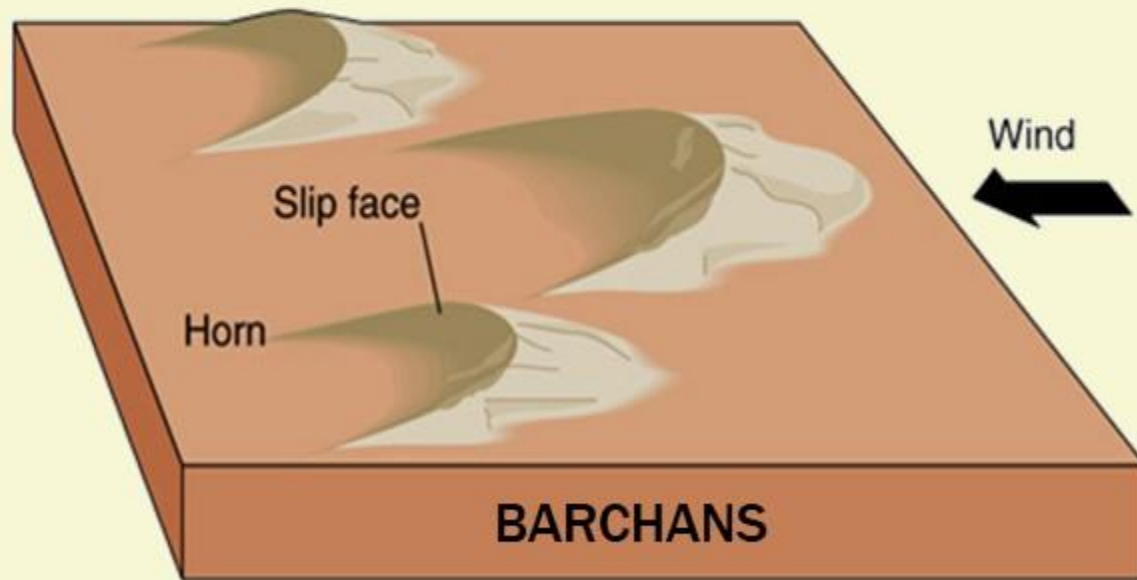
# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

2

### BARCHANS

2 Parabolic dunes form when sandy surfaces are partially covered with vegetation.



*That means parabolic dunes are reversed barchans with wind direction being the same.*



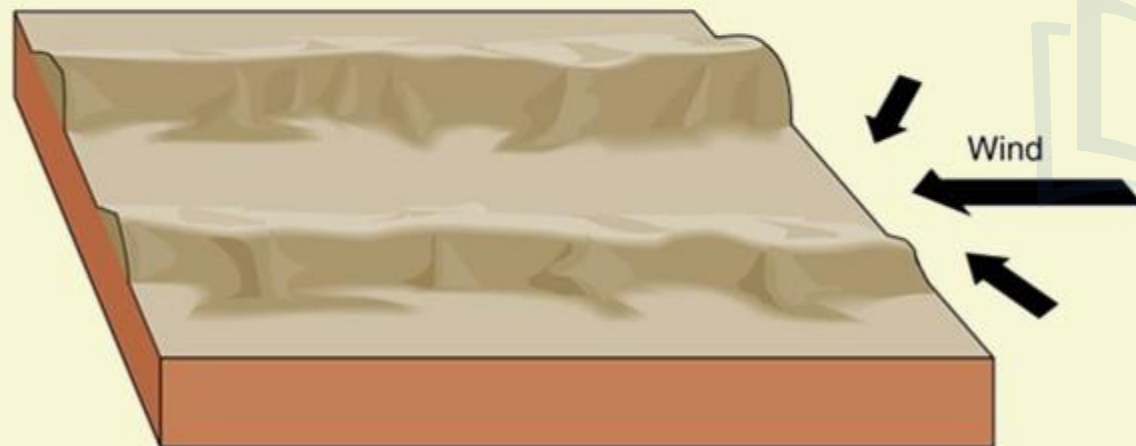
# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

2

### BARCHANS

3 Seif is similar to barchans with a small difference.



**LONGTITUDINAL DUNES OR SEIFS**

4 Seif has only one wing or point. This happens when there is shift in wind conditions.

5 The lone wings of seifs can grow very long and high.

6 Longitudinal dunes form when supply of sand is poor and wind direction is constant.

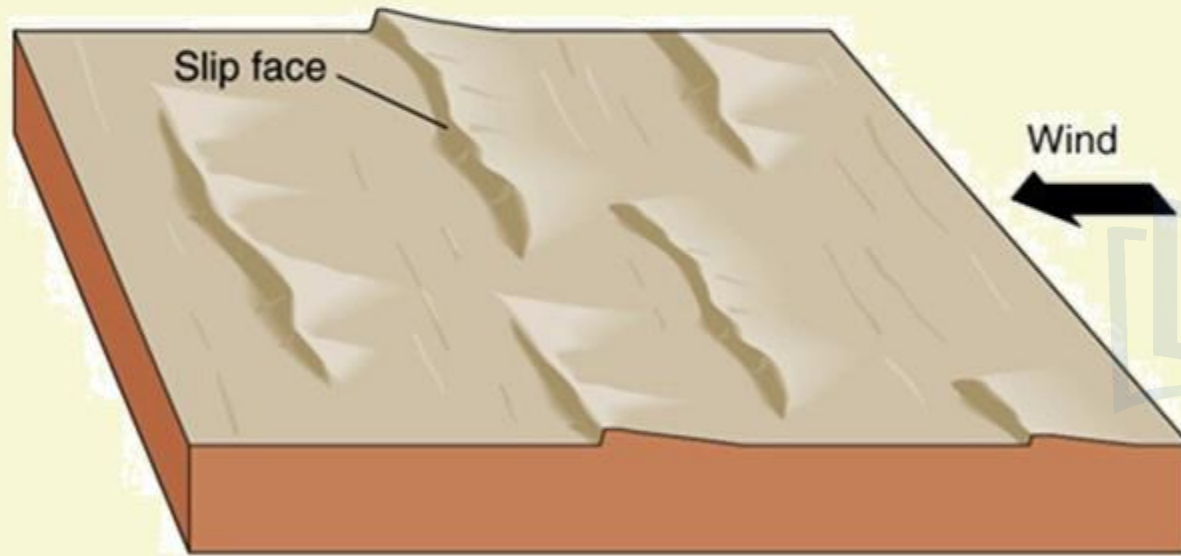
7 They appear as long ridges of considerable length but low in height.

# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

2

### BARCHANS



**TRANSVERSE DUNES**

**8** Transverse dunes are aligned perpendicular to wind direction.

**9** These dunes form when the wind direction is constant. It is an elongated feature at right angles to the wind direction. They may be very long and low in height.



# ARID TOPOGRAPHY

## DEPOSITIONAL LANDFORMS

2

### BARCHANS



9

When sand is plenty, quite often, the regular shaped dunes coalesce and lose their individual characteristics.

10

Most of the dunes in the deserts shift and a few of them will get stabilized especially near human habitations.

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[suggestions@learningspace.in](mailto:suggestions@learningspace.in)

To Contact us:

[info@learningspace.in](mailto:info@learningspace.in)

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K. Srikanth  
D. Sunil Kumar  
L. V. Krishna  
Y. Deepthi

Amrita Naidu  
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